Exam Summary

Module Name

Student

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Part 1 — The Building Blocks

Type	Dims	Notation	Example
Scalar	0D	a	single loss value
Vector	1D	$\boldsymbol{x}$ — always column	$[1, 2, 3]^\top$
Matrix	2D	$\boldsymbol{A}$, element $A_{i,j}$	weight layer
Tensor	3D+	$\boldsymbol{A}$, element $A_{i,j,k}$	image batch

Column vector and matrix notation:

$$\boldsymbol{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} \quad \boldsymbol{A} = \begin{pmatrix} A_{1,1} & A_{1,2} \\ A_{2,1} & A_{2,2} \\ A_{3,1} & A_{3,2} \end{pmatrix}$$

Part 2 — Operations

Matrix addition (same size only):

- *Used for:* combining two sets of features, adding residuals in ResNets

$$C_{i,j} = A_{i,j} + B_{i,j}$$

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} + \begin{pmatrix} 5 & 6 \\ 7 & 8 \end{pmatrix} = \begin{pmatrix} 6 & 8 \\ 10 & 12 \end{pmatrix}$$

Scalar on matrix (a = scale, c = bias/shift):

- *Used for:* normalising data, applying a weight + bias to a whole layer

$$D_{i,j} = a \cdot B_{i,j} + c$$

$$2 \cdot \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} + 10 = \begin{pmatrix} 12 & 14 \\ 16 & 18 \end{pmatrix} \quad \rightarrow \text{every element } \times 2\text{, then } +10$$

Broadcasting (vector added to every column of matrix):

- *Used for:* adding a bias vector to an entire batch of inputs at once

$$C_{i,j} = A_{i,j} + b_j$$

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{pmatrix} + \begin{pmatrix} 10 \\ 20 \end{pmatrix} = \begin{pmatrix} 11 & 22 \\ 13 & 24 \\ 15 & 26 \end{pmatrix}$$

Matrix multiplication:

- *Used for:* the core forward pass — transforming inputs through a weight layer: $\boldsymbol{y} = W\boldsymbol{x}$

$$\boldsymbol{C} = \boldsymbol{AB} \quad C_{i,j} = \sum_k A_{i,k} \cdot B_{k,j}$$

Shape: $(m \times k) \cdot (k \times n) = (m \times n)$ — inner dims must match.

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 5 & 6 \\ 7 & 8 \end{pmatrix} = \begin{pmatrix} 19 & 22 \\ 43 & 50 \end{pmatrix}$$

Inner / dot product (vectors only, result is a scalar):

- *Used for:* measuring similarity between two vectors; attention scores in Transformers

$$\boldsymbol{x}^\top \boldsymbol{y} = \sum_i x_i y_i = \|\boldsymbol{x}\|_2 \|\boldsymbol{y}\|_2 \cos\theta$$

$$(1 \;\; 2 \;\; 3) \begin{pmatrix} 4 \\ 5 \\ 6 \end{pmatrix} = 1 \cdot 4 + 2 \cdot 5 + 3 \cdot 6 = 32$$

Hadamard (element-wise) product:

- *Used for:* gating mechanisms (LSTM, GRU), applying masks, dropout

$$\boldsymbol{C} = \boldsymbol{A} \odot \boldsymbol{B} \quad C_{i,j} = A_{i,j} \cdot B_{i,j}$$

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \odot \begin{pmatrix} 5 & 6 \\ 7 & 8 \end{pmatrix} = \begin{pmatrix} 5 & 12 \\ 21 & 32 \end{pmatrix}$$

Properties of matrix multiplication:

- Distributive: $A(B + C) = AB + AC$
- Associative: $A(BC) = (AB)C$

Exam Trap: $AB \neq BA$ — matrix multiplication is **NOT commutative**. This is on every exam.

Transpose:

- *Used for:* converting column↔row vectors, computing dot products ($\boldsymbol{x}^\top \boldsymbol{y}$), flipping weight matrices in backpropagation

$$\left(A^\top\right)_{i,j} = A_{j,i} \quad (AB)^\top = B^\top A^\top$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}^\top = \begin{pmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{pmatrix} \quad \rightarrow \text{(2 times 3) becomes (3 times 2)}$$

Part 3 — Norms

- *Used for:* measuring magnitude of weights, gradients, errors; regularisation (L1/L2); comparing vectors

General L^p norm:

$$\|\boldsymbol{x}\|_p = \left(\sum_i |x_i|^p\right)^{\frac{1}{p}}, \quad p \geq 1$$

Example with $\boldsymbol{x} = [3, -4]^\top$:

$$\|\boldsymbol{x}\|_1 = 3 + 4 = 7 \quad \|\boldsymbol{x}\|_2 = \sqrt{9 + 16} = 5 \quad \|\boldsymbol{x}\|_\infty = 4$$

Norm	**Formula**	**Use**
L^2 (Euclidean)	$\|\boldsymbol{x}\|_2 = \sqrt{\sum_i x_i^2}$	most common
Squared L^2	$\|\boldsymbol{x}\|_2^2 = \boldsymbol{x}^\top \boldsymbol{x}$	easier to compute
L^1	$\|\boldsymbol{x}\|_1 = \sum_i \lvert x_i \rvert$	sparsity
L^0	count of non-zeros	**not a true norm**
L^∞	$\|\boldsymbol{x}\|_\infty = \max_i \lvert x_i \rvert$	max element
Frobenius	$\|A\|_F = \sqrt{\sum_{i,j} A_{i,j}^2} = \sqrt{\operatorname{tr}(AA^\top)}$	matrix size

Part 4 — Identity, Inverse, Solving Systems

- *Used for:* solving systems of equations $A\boldsymbol{x} = \boldsymbol{b}$; finding model parameters given data

$$A^{-1}A = I \quad A\boldsymbol{x} = \boldsymbol{b} \Rightarrow \boldsymbol{x} = A^{-1}\boldsymbol{b}$$

Example — solve $A\boldsymbol{x} = \boldsymbol{b}$:

$$A = \begin{pmatrix} 2 & 0 \\ 0 & 4 \end{pmatrix} \quad \boldsymbol{b} = \begin{pmatrix} 6 \\ 8 \end{pmatrix} \quad \to A^{-1} = \begin{pmatrix} 1/2 & 0 \\ 0 & 1/4 \end{pmatrix} \quad \to \boldsymbol{x} = \begin{pmatrix} 3 \\ 2 \end{pmatrix}$$

A^{-1} exists only if: A is square AND columns are linearly independent (non-singular).

Situation	**Solutions**
$m > n$ (overdetermined)	None (generically)
$m < n$ (underdetermined)	Infinitely many
$m = n$, independent cols	**Exactly one**

Part 5 — Special Matrices

Diagonal:

- *Used for:* efficient scaling; eigenvalue matrices in decompositions; fast inversion

$$\operatorname{diag}(\boldsymbol{v})\boldsymbol{x} = \boldsymbol{v} \odot \boldsymbol{x} \quad \operatorname{diag}(\boldsymbol{v})^{-1} = \operatorname{diag}\left([1/v_1, ..., 1/v_n]^\top\right)$$

$$\operatorname{diag}\left(\begin{pmatrix} 2 \\ 3 \\ 4 \end{pmatrix}\right)\boldsymbol{x} = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 4 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 2x_1 \\ 3x_2 \\ 4x_3 \end{pmatrix} \quad \to \text{same as } \begin{pmatrix} 2 \\ 3 \\ 4 \end{pmatrix} \odot \boldsymbol{x}$$

Symmetric:

- *Used for:* covariance matrices, distance matrices, any matrix of pairwise values

$$A = A^\top$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 5 & 4 \\ 3 & 4 & 6 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 5 & 4 \\ 3 & 4 & 6 \end{pmatrix}^\top \quad \to \text{mirrored across diagonal}$$

Orthogonal (rows AND cols are orthonormal):

- *Used for:* rotation matrices, V and U in SVD, efficient inversion (just transpose)

$$A^\top A = AA^\top = I \quad \Rightarrow \quad A^{-1} = A^\top$$

Inverting an orthogonal matrix = just transposing it — no computation needed.

Part 6 — Eigendecomposition

- *Used for:* understanding what a matrix does geometrically; PCA; finding principal directions of data; quadratic optimisation

$$A\boldsymbol{v} = \lambda \boldsymbol{v}$$

- $\boldsymbol{v}$ = eigenvector (direction unchanged by A, unit length $\|\boldsymbol{v}\| = 1$)
- λ = eigenvalue (scaling factor along that direction)

Example — $A = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}$ has eigenvalues $\lambda_1 = 4, \lambda_2 = 2$:

$$A\begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 4 \\ 4 \end{pmatrix} = 4\begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad \to \text{direction } [1,1]^\top \text{ scaled by } 4$$

$$A\begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 2 \\ -2 \end{pmatrix} = 2\begin{pmatrix} 1 \\ -1 \end{pmatrix} \quad \to \text{direction } [1,-1]^\top \text{ scaled by } 2$$

Full decomposition:

$$A = V \operatorname{diag}(\boldsymbol{\lambda}) V^{-1} \quad V = \left[\boldsymbol{v}^{(1)}, \dots, \boldsymbol{v}^{(n)}\right]$$

Intuition: **rotate** (by V) → **scale** (by λ) → **rotate back** (V^{-1}).

For real symmetric matrices: n real eigenvalues, orthogonal eigenvectors → $V^{-1} = V^\top$.

Optimization (exam!):

$$\max_{\|\boldsymbol{x}\|=1} \boldsymbol{x}^\top A \boldsymbol{x} \to \boldsymbol{x} = \text{eigenvec of max eigenvalue}$$

$$\min_{\|\boldsymbol{x}\|=1} \boldsymbol{x}^\top A \boldsymbol{x} \to \boldsymbol{x} = \text{eigenvec of min eigenvalue}$$

Part 7 — Singular Value Decomposition (SVD)

- *Used for:* pseudoinverse, data compression, dimensionality reduction (PCA), recommender systems, solving any linear system

Works for **any** matrix (generalization of eigendecomposition):

$$A = UDV^\top$$

- U: orthogonal — left singular vectors (output directions)
- D: diagonal — singular values (how much each direction is stretched)

- V: orthogonal — right singular vectors (input directions)

Example — $A = \begin{pmatrix} 3 & 0 \\ 0 & 2 \end{pmatrix}$: already diagonal, so $U = I$, $D = A$, $V = I$. For a general matrix the singular values in D tell you how much the matrix stretches space in each direction — largest first.

Part 8 — Moore-Penrose Pseudoinverse

- *Used for:* least-squares regression, solving overdetermined systems (more equations than unknowns); any time a true inverse doesn't exist

When A^{-1} doesn't exist:

$$A^{+} = VD^{+}U^{\top} \quad (D^{+} : \text{reciprocal of each non-zero diagonal of D})$$

Example — $D = \begin{pmatrix} 3 & 0 \\ 0 & 2 \\ 0 & 0 \end{pmatrix} \to D^{+} = \begin{pmatrix} 1/3 & 0 & 0 \\ 0 & 1/2 & 0 \end{pmatrix}$

Case	Result
More rows than cols (overdetermined)	Minimises $\|A\boldsymbol{x} - \boldsymbol{b}\|_2$ (least squares)
More cols than rows (underdetermined)	Minimises $\|\boldsymbol{x}\|_2$ among all solutions

Part 9 — Trace

- *Used for:* computing Frobenius norm, simplifying matrix derivative expressions, checking eigenvalue sums

$$\text{tr}(A) = \sum_i A_{i,i} = \sum_i \lambda_i$$

Example:

$$\text{tr}\left(\begin{pmatrix} 1 & 9 & 7 \\ 2 & 5 & 3 \\ 4 & 6 & 8 \end{pmatrix}\right) = 1 + 5 + 8 = 14$$

- $\text{tr}(A^{\top}) = \text{tr}(A)$
- $\text{tr}(ABC) = \text{tr}(CAB) = \text{tr}(BCA) \leftarrow$ cyclic invariant
- $\|A\|_F = \sqrt{\text{tr}(AA^{\top})}$

Part 10 — Determinant

- *Used for:* checking if a matrix is invertible; measuring volume scaling; appears in probability density formulas (multivariate Gaussian)

$$\det(A) = \prod_i \lambda_i$$

Example:

$$\det\left(\begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}\right) = 3 \cdot 3 - 1 \cdot 1 = 8 \quad \to \lambda_1 \cdot \lambda_2 = 4 \cdot 2 = 8 \checkmark$$

- $\det(A) = 0 \to$ singular, not invertible (at least one eigenvalue is zero)
- $\det(A) \neq 0 \to$ invertible

- Geometric meaning: factor by which A scales volume

Cheat Sheet

WEEK 1 — LINEAR ALGEBRA CHEAT SHEET

Structures: scalar=0D · vector=1D(col) · matrix=2D · tensor=3D+

Multiply:

- Standard: $C_{i,j} = \sum_k A_{i,k} B_{k,j}$
- Dot product: $\boldsymbol{x}^\top \boldsymbol{y} = \|\boldsymbol{x}\| \|\boldsymbol{y}\| \cos\theta$
- Hadamard $\odot$: $C_{i,j} = A_{i,j} B_{i,j}$
- $AB \neq BA$ — NEVER commutative

Norms: $L^2 = \sqrt{\sum x^2}$ · $L^1 = \sum |x|$ · $L^\infty = \max |x|$ · $\|A\|_F = \sqrt{\mathrm{tr}(AA^\top)}$

Special: Symmetric $A = A^\top$ · Orthogonal $A^{-1} = A^\top$ · Diagonal: invert = reciprocal diagonal

Eigen: $A\boldsymbol{v} = \lambda \boldsymbol{v} \to A = V \,\mathrm{diag}(\lambda) V^{-1}$

SVD: $A = UDV^\top$ (always works)

Pseudoinverse: $A^+ = VD^+U^\top$

Trace: $\mathrm{tr}(A) = \sum A_{i,i} = \sum \lambda_i$ (cyclic invariant)

Det: $\det(A) = \prod \lambda_i$ · zero = singular = not invertible